

| Question |     |      | Marking details                                                                                                                                                                                                                            |          | Marks available |          |           |          |           |  |
|----------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|----------|-----------|----------|-----------|--|
|          |     |      |                                                                                                                                                                                                                                            | AO1      | AO2             | AO3      | Total     | Maths    | Prac      |  |
| 4        | (a) |      | Valid method for obtaining gradient (implied if correct) (1)<br>Max = ~0.021 or min = ~0.017 (1) [OK to give 21.0 & 17.5 mT A <sup>-1</sup> ]<br>Mean correct (~0.019) (1)<br>Uncertainty = 8.5 % – 11% (1) [ no s.f. penalty]             |          | 4               |          | 4         | 4        | 4         |  |
|          | (b) |      | Gradient = $\mu_0 n$ used or implied ( $n$ =15 200) (1)<br>$N$ = 3 780 (1)<br>$N$ = 3 800 ± 400 (1) [allow 1 or 2 s.f. in uncertainty 1 more s.f. in $N$ ]                                                                                 |          |                 | 3        | 3         | 3        | 3         |  |
|          | (c) | (i)  | 5 000 is outside uncertainty range [or equiv.] (1)<br>Straight line (1)<br>Through all error bars (1)<br>Lines straddle the origin (1) [allow ....through origin]                                                                          |          |                 | 4        | 4         |          | 4         |  |
|          |     | (ii) | Flux density least accurate/probe not in centre [end effect] /not aligned [90° to axis](1)<br>Get better Hall probe with higher [smaller / better] resolution / put in centre/align properly (1) [not: repeat readings / better apparatus] |          |                 | 2        | 2         |          | 2         |  |
|          |     |      | <b>Question 4 total</b>                                                                                                                                                                                                                    | <b>0</b> | <b>4</b>        | <b>9</b> | <b>13</b> | <b>7</b> | <b>13</b> |  |